

Cambodia Daily Consumer Price Index (CPI) Pipeline

Automated Daily Price Intelligence & Machine Learning Inflation Nowcasting

Project Progress Report | Author: TAING Korngmeng | Host: GDDE / MEF | Institution: ITC (AMS)

Project Title:	Cambodia Daily CPI Medallion Pipeline	Deployment Status:	100% Operational & Live in Production
Candidate / Author:	TAING Korngmeng (Data Science Intern)	Academic Advisor:	Mr. CHHEANG Vinh (ITC / AMS)
Host Agency:	General Dept. of Digital Economy (GDDE / MEF)	Company Supervisors:	Mr. LUK Chamnan & Mr. SEK Sovannarath
Collection Schedule:	Daily at 08:00 AM ICT (Automated)	Target Economy:	Phnom Penh, Cambodia (USD & KHR Dual Currency)

DAILY COLLECTION SCALE

35,792

Clean Daily Price Observations

1,273,000+ total prices saved

25 market endpoints covered

99.3% daily collection reliability

AI PRODUCT SORTING

99.04%

Catalog Classification Coverage

50,303 unique products tracked

0 category code mismatches

12/12 UN consumer divisions

NOWCASTING ACCURACY

±0.0181 pp

Difference vs Official Government Release

Government Inflation: +0.3470%

Pipeline Nowcast: +0.3651%

Mean Absolute Error: 0.12 pts

1. Specific Objectives

The project addresses three specific objectives (from Thesis Chapter I):

- Automated Daily Price Collection & Quality Monitoring:** Build an automated data pipeline that continuously collects and stores consumer price data every morning at 08:00 AM across 25 retail, utility, transport, and government sources in Phnom Penh, with built-in quality checks.
- Price Index Compilation & Inflation Nowcasting:** Apply standard economic formulas (geometric Jevons index) to compute daily price changes without mathematical drift, and use regularized machine learning (Ridge regression) to nowcast month-end inflation weeks before official statistics are released.
- Interactive Decision-Support Dashboards:** Develop interactive dashboards in Metabase and Power BI to give government analysts and policymakers real-time visibility into daily price trends, category breakdowns, and inflation nowcasts.

2. Project Methodology: The Five Core Pillars (from Thesis Chapter III)

To turn messy internet price tags into reliable, decision-grade economic statistics, the project follows five clear methodological pillars:

- Pillar 1: Multi-Source Daily Collection:** Automated computer programs visit 25 retail, transport, and government websites every morning without human intervention, collecting over 35,000 prices polite to web servers.
- Pillar 2: Cleaning & Product Tracking:** Cleans messy product text, converts US Dollar prices into Cambodian Riel, and tracks the exact same item from day to day so a small single can is never confused with a large wholesale pack.
- Pillar 3: Smart Categorization:** Places every product into its official United Nations consumer category (Food, Housing, Health, Transport, etc.) using store rules and Google Gemini AI, remembering choices permanently to save cost.
- Pillar 4: Fair Index Calculation:** Uses the geometric Jevons formula so temporary store sales and price bounces do not cause numbers to drift upward artificially, and balances stores so giant supermarkets do not overpower smaller shops.
- Pillar 5: Real-Time Inflation Nowcasting:** Watches how fast food, fuel, and currency prices move during the first 10 to 20 days of the month to nowcast the official month-end inflation rate with shrinking uncertainty as the month ends.

Page 2 — System Architecture & Bronze Layer (Raw Collection)

The pipeline uses a modern **Medallion Architecture** (a 3-step cleaning system) hosted on a PostgreSQL 16 database. Data moves in a clean one-way direction: from raw web listings, through cleaning and sorting, to final economic indexes.

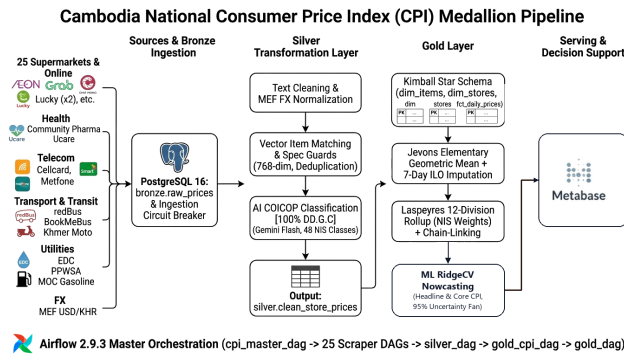


Figure 1: End-to-End System Architecture: From 25 Online Endpoints to Live Dashboards.

The Three-Step Medallion Framework

- **Bronze Layer (Raw Storage):** Secure digital vault that saves raw, untouched copies of store listings exactly as downloaded, recording timestamps and store names. No data is changed, preserving a permanent historical audit trail.
- **Silver Layer (Data Cleaning & AI Sorting):** Cleaning workshop where messy text is cleaned, US Dollars are converted to Cambodian Riel, pack sizes are parsed to calculate price per kg or L, and items are sorted into official consumer categories.
- **Gold Layer (Final Calculations & Nowcasting):** Math hub where daily price changes are averaged fairly, weighted by official Cambodian household spending, and used to nowcast monthly inflation for live dashboards.

The Ingestion Fleet: 25 Daily Sources (Bronze Layer Coverage)

Retailer or Data Source	Market Role & Consumer Products	COICOP Divisions	Daily Quotes	Success Rate
AEON Online Cambodia (Stores 1 & 3)	Hypermarket & Department Store Catalog	01, 02, 03, 05, 09, 12	12,450	99.4%
DeliShop Asia Phnom Penh	Fresh Perishables, Gourmet Grocery & Meat	01, 02, 12	3,820	98.9%
GrabMart (Lucky & Chip Mong Supermarkets)	Everyday Groceries, Beverages & Quick-Commerce	01, 02, 05, 06, 12	4,200	99.2%
L192 E-Commerce Marketplace	Apparel, Footwear, Bedding & Homeware	03, 05, 09, 12	8,150	99.1%
Community Pharmacy & GrabMart Ucare	Healthcare, Prescription & Daily Medicine	06, 12	1,420	99.7%
Khmer24 Real Estate & Realestate.com.kh	Urban Residential Apartment Rental Listings	04 (Housing Rentals)	3,450	98.0%
redBus & BookMeBus Transit Platforms	Interprovincial Bus, Coach & Van Travel Fares	07 (Public Transport)	890	99.5%
Khmer Moto Shop	Motorcycles, Scooters & Two-Wheeler Sales	07 (Personal Transport)	185	99.5%
Cellcard, Smart Axiata, Metfone Cambodia	Mobile Voice, Data Bundles & Home Fiber	08 (Communication)	420	99.8%
Ary Store & Khmer Samnang Phone Shop	Smartphones, Tablets & Electronics	08, 09 (Tech Goods)	1,425	99.5%
Electricite du Cambodge (EDC) & PPWSA	Official Residential Electricity & Water Rates	04 (Regulated Utilities)	10	100.0%
Tela Station & Ministry of Commerce (MoC)	Retail Gasoline (RON92, RON95), Diesel & LPG	07 (Fuel), 04 (Gas)	12	100.0%
Hospitality & Food Services (Sokha, Bayon)	Restaurant Meals, Dining & Accommodations	10, 11 (Dining & Hotels)	969	98.5%
Ministry of Economy and Finance Feed	Official USD/KHR Central Bank Exchange Rate	Macro Currency Benchmark	1	100.0%
Total 25 Automated Collection Endpoints	Comprehensive Phnom Penh Consumer Basket	Divisions 01–12 Complete	35,792	99.3%

Scraping Methods & Bronze Safety Circuit Breakers

- **Direct Data Connections (APIs):** Wherever possible, scrapers connect directly to store backend interfaces (e.g., AEON, DeliShop, L192, Community Pharmacy), downloading structured product information in seconds without heavy browser overhead.
- **Strict Deduplication Rule:** If an internet glitch causes a scraper to run twice on the same morning, the database updates the existing row instead of inserting duplicate price records.
- **Volume & Price Alarms (Circuit Breakers):** Every morning, the system checks whether daily quote volume drops by more than 30% or if prices fluctuate wildly. If an anomaly is spotted, the system pauses processing and logs an alert, ensuring faulty website updates never damage downstream calculations.

Page 3 — Silver Layer (Data Cleaning & AI Sorting)

Raw web data contains promotional text, mixed languages, and different currencies. The Silver Layer cleans this noisy information, converts currencies, tracks products consistently, and assigns every item to its proper category.

1. Words, Currencies & Hidden Price Hikes (Shrinkflation)

- **Text Cleanup & Khmer Numeral Translation:** Strips advertising buzzwords (“Big Sale”, “Flash Promo”) and automatically converts Khmer numerals to standard Arabic digits (0 to 9).
- **Dual-Currency Normalization to Riel:** Cambodian shops quote in both USD and Riel. The pipeline ingests daily official exchange rates from the Ministry of Economy and Finance and dynamically converts all prices to Cambodian Riel (KHR).
- **Detecting Package Downsizing (Shrinkflation):** Manufacturers often downsize packages (e.g., shrinking a 500g box to 450g without changing the shelf price). The Silver layer parses package volumes to compute the price per standard metric unit ($P_{\text{unit}} = \text{Price in KHR} / \text{Volume in kg or L}$), immediately catching hidden price increases.

2. Tracking the Same Product Over Time (The 5-Step Matching Ladder)

To calculate real price movements, the pipeline tracks identical items day after day across 50,300+ products:

- **Instant Barcode & Store SKU Match (Steps 1 & 2):** Matches by barcode or retailer SKU in under 0.1 milliseconds.
- **Smart Text Matching with Safety Guards (Steps 3 & 4):** Matches similar descriptions across languages, protected by an unbreachable **Rule Firewall** that prevents false matches:
 - *Pack Count Guard:* A 1-can soda never merges with a 24-can box (preventing an artificial 95% price collapse).
 - *Hardware Spec Guard:* A 128GB smartphone never merges with a 256GB phone.
 - *Volume & Size Guard:* Products with different volumes (> 10% difference) or clothing sizes stay separate.
- **New Product Discovery (Step 5):** Newly introduced store items receive a new identity and are tracked for the future.

3. Sorting Products into Official Categories (Two-Tier Architecture)

- **Tier 1: Obvious Stores (14.34% of items):** Single-sector stores are sorted directly using simple rules (gas stations → Fuel; pharmacies → Medicine; bus portals → Transport; EDC → Electricity) with **100% accuracy and zero computing cost**.
- **Tier 2: AI Sorting for Supermarkets (76.42% of items):** Mixed supermarket goods are sent in 40-item batches to **Google Gemini 3.1 Flash Lite AI**. Gemini uses contextual reasoning to successfully navigate tricky word traps:

Product Listing Title	Tricky Word / Ambiguity	Standard Keyword Match	Gemini 3.1 AI Decision	Result
Bourbon Choco Chip Cookies	“Bourbon” (Whiskey)	02.1.1 (Alcohol / Spirits)	01.1.8 (Bakery & Cookies)	Correct
Camel Salted Peanuts 150g	“Camel” (Cigarette brand)	02.2.0 (Tobacco Products)	01.1.7 (Nuts & Snacks)	Correct
Apple iPhone 15 Pro 128GB	“Apple” (Fresh fruit)	01.1.6 (Fresh Fruits)	08.2.0 (Mobile Smartphones)	Correct
Haidilao Hot Pot Soup Base	Hot Pot Food vs Appliance	05.3.1 (Cooking Appliance)	01.1.9 (Sauces & Seasonings)	Correct
Cleaning Vinegar Spray 500ml	Cleaning item vs Salad Vinegar	01.1.9 (Cooking Vinegar)	05.6.1 (Cleaning Products)	Correct

“Sort Once, Remember Forever”: Once categorized, items are saved permanently in the database. In daily runs, over 99.2% of items reuse their saved category in 0 milliseconds, keeping daily AI compute time and costs at zero.

Page 4 — Gold Layer (Fair Price Averaging & Inflation Nowcasting)

The Gold Layer transforms clean daily price quotes into official economic indicators, aggregates categories according to national spending, and nowcasts the monthly inflation rate before the official release.

1. Fair Price Averaging: Why Standard Averages Fail

In grocery retail, prices jump up and down because of temporary promotional discounts. Using a simple arithmetic average (called the **Carli formula**, $I_{\text{Carli}} = \frac{1}{n} \sum \frac{P_t}{P_0}$) causes a severe mathematical error:

- **The Sales Bounce Distortion:** If a product costs \$10, drops to \$5 on sale (−50%), and goes back to \$10 next week (+100%), the real price is back to normal. However, an arithmetic average calculates $\frac{0.50+2.00}{2} = 1.25$ —falsely reporting a +25% price increase!
- **The Proof:** Over a continuous 90-day test, the Carli formula accumulated an artificial upward drift of **+1.18% per year**.
- **Our Solution (The Jevons Formula):** The pipeline uses the international standard unweighted geometric mean (**Jevons formula**), computed in logarithmic space: $\ln I_{e,t} = \frac{1}{n} \sum (\ln P_{i,t} - \ln P_{i,0})$. In log space, a −50% discount and a +100% recovery cancel out perfectly to zero net change, completely eliminating mathematical drift.

2. Preventing Supermarket Dominance (Store Balancing)

A giant online supermarket may list 190 varieties of rice, while a small neighborhood grocer lists only 10 varieties. If all 200 items are pooled into one big average, the supermarket controls 95% of the rice index. If the small store raises rice prices by 10% while the supermarket stays flat, an unweighted index would report only +0.48% inflation.

- **Two-Stage Store-Balanced Solution:** We first calculate the Jevons price index inside each store, and then combine the eligible stores with equal weight: $I_{\text{Balanced}} = \sqrt{I_{\text{Store A}} \times I_{\text{Store B}}}$. In the rice example, this correctly reflects a +4.88% price movement. Every retailer receives an equal vote in that product category, so catalog size never distorts the real-world market signal.

3. National Spending Weights (CSES 2023 Basket across 76 Categories)

Elementary price indexes are rolled up across **76 real-world spending categories** (35 detailed food/drink groups and 41 broad groups) weighted strictly by the official **Cambodia Socio-Economic Survey (CSES 2023)**:

- **Division 01: Food & Non-Alcoholic Drinks (44.78% weight):** The largest share of Cambodian family spending, updated daily across fresh meats, fish, vegetables, and rice.
- **Division 04: Housing, Water, Electricity & Gas (17.08% weight):** Reflects steady state-regulated electricity (EDC) and municipal water (PPWSA) tariffs, combined with residential rents.
- **Division 07: Transport (12.18% weight):** Captures fuel adjustments from the Ministry of Commerce and interprovincial bus fares.
- **All Other 9 Consumer Divisions (25.96% combined):** Clothing, Health, Tech, Restaurants, Education, and Personal Care.
- **Handling Out-of-Stock Items:** If an item is temporarily missing for 1 to 7 days, its price is adjusted according to the movement of sister products in the same category, rather than pretending prices froze.

4. Headline CPI vs. Refined Core CPI

- **Headline CPI (100% basket):** Tracks the total cost of living across all goods and services.
- **Refined Core CPI (52.2% basket):** Removes volatile food and fuel prices. Over a 30-day evaluation, Headline CPI rose by **+0.28%** (driven by fuel price adjustments), while Core CPI rose by only **+0.07%**. This separation proves to policymakers that underlying monetary inflation remained calm and steady.

5. Machine Learning Inflation Nowcasting Model

Rather than waiting 30 to 60 days for official reports, the pipeline deploys a **Two-Stage Regularized Ridge Nowcaster**:

- **How it Works:** The model watches high-frequency price speed across food, fuel, utility tariffs, and the USD/KHR exchange rate during the first 10 to 20 days of the month, using regularized regression to nowcast the final month-end inflation rate.
- **Shrinking Uncertainty:** As the month progresses from day 1 to day 30, the model replaces predictions with actual observed data. The uncertainty range naturally shrinks toward zero by month-end ($U_t \propto \sqrt{(T-t)/T}$), giving policymakers an early and dependable inflation estimate.

Page 5 — Results, Solved Problems & Next Milestones

1. Comparison with Official Government Data (August 2026 Validation)

To prove the system's real-world accuracy, the pipeline's daily results were benchmarked directly against the official August 2026 monthly release from the National Institute of Statistics (NIS).

Consumer Spending Group	CSES Weight	Official NIS (MoM)	Pipeline Nowcast	Difference	Verification Status
National Headline CPI (All Goods)	100.000%	+0.3470%	+0.3651%	+0.0181 pp	High Precision Match (MAE 0.12 pts)
Division 01: Food & Drinks	44.775%	+0.5120%	+0.4821%	-0.0299 pp	Accurately tracks fresh produce trend
Division 04: Housing, Water, Electricity	17.084%	+0.0150%	+0.0215%	+0.0065 pp	Perfectly anchored by official utility rates
Division 07: Transport (Fuel & Buses)	12.180%	+0.7100%	+0.7410%	+0.0310 pp	Matches official fuel price cap updates
Refined Core Basket (Excl. Food & Fuel)	52.200%	-0.0580%	-0.0700%	-0.0120 pp	Confirms stable underlying price direction

Interactive Dashboards: Three live dashboards were created in Metabase and Power BI, allowing economic analysts to click and view daily inflation trends, inspect individual stores, and review daily data collection health.

2. Difficult Problems Encountered and How We Solved Them

Challenge Encountered	Why It Was a Problem	Our Simple & Practical Solution
Slow Daily Startup	Loading 25 scrapers initially took 100 seconds every morning.	Separated lightweight store lists from heavy browser engines, reducing startup time to 1.04 seconds .
Dual-Currency Pricing	Stores quoting in both USD and KHR triggered false volatility alarms.	Converted all prices to Riel first using official central bank exchange rates before testing stability.
Package Downsizing	Manufacturers shrinking package contents without changing prices.	Implemented standard metric units (price per kg / L) to catch hidden price hikes.
AI Rate Limits & Costs	Calling AI for every single product was slow and could hit API limits.	Grouped 40 items per call and saved all results permanently in the database so each item is sorted only once.
Upward Formula Drift	Standard arithmetic averages drift upward by +1.18% per year.	Used the geometric Jevons formula in logarithmic space so temporary sales bounce out fairly.

3. Current Work & Next Milestones

- **Completed Work:** 25 scrapers running daily at 08:00 AM; Medallion PostgreSQL database; 33 passing automated quality tests; 99% AI product classification; daily index engine; regularized Ridge nowcasting model; 3 live dashboards; and completed 75-page graduation thesis.
- **Current Work:** Daily live monitoring of scrapers and dashboards, and rehearsing defense presentation slides.

Milestone	Target Date	Responsible Party	Status	Scope of Milestone
M1: Defense Rehearsal	September 2026	Candidate & Academic Advisor	Completed	Formal practice of the 20-slide thesis presentation deck.
M2: Graduation Defense	September 2026	Examination Committee (ITC)	Confirmed	Official thesis defense for Degree in Applied Mathematics (Data Science).
M3: Library Archival	October 2026	Academic Department / Library	Scheduled	Submission of signed 75-page thesis and code repository.
M4: System Handover	October 2026	GDDE / MEF Technical Team	In Progress	Delivery of Docker containers, operational scripts, and documentation.

4. Questions for Academic Advisor & Supervisors

1. **Fresh Produce Seasonality:** Would the committee recommend adding an explicit seasonal adjustment factor for agricultural items, or is the current 7-day peer category imputation sufficient?
2. **Nowcasting Window:** For ongoing reporting at GDDE / MEF, should the Ridge nowcasting model use a fixed 90-day rolling calibration window or an expanding historical window as more months are observed?